

Project Sprint #1

The Solitaire game is described in CS449HomeworkOverview. You should read the description very carefully.

In this assignment, you aim to specify the requirements (i.e., user stories and acceptance criteria) of the target software that allows a human player to play a Solitaire game. These requirements will be fully implemented by the end of sprint 3. The minimum features include **choosing the board size and type, starting a new game, making a move, determining if a game is over**. The following is a sample GUI layout.

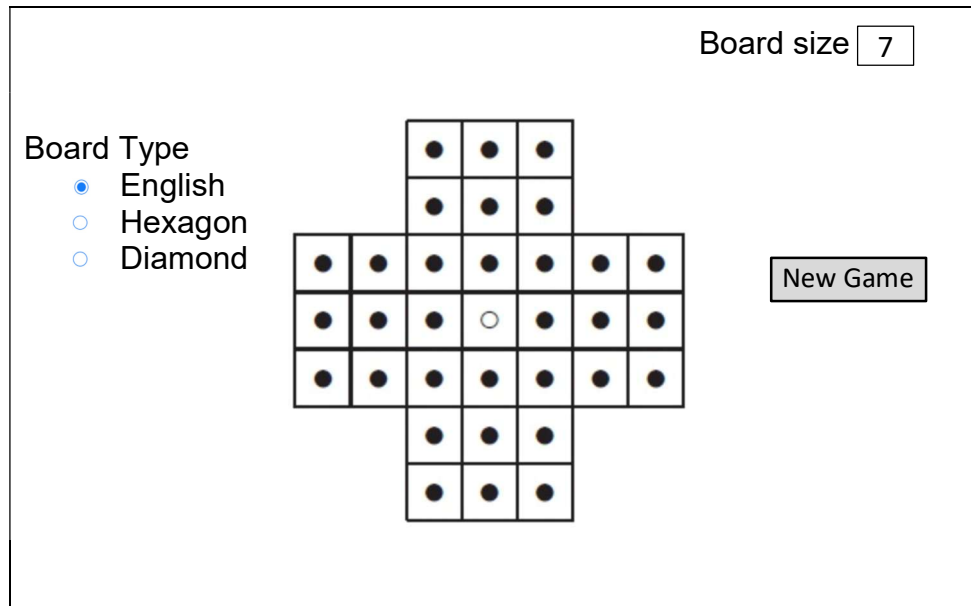


Figure 1. Sample GUI layout of the first working program by the end of Sprint 3

Use the following tables to document your user stories and acceptance criteria.

You are required to use the free ChatGPT or another LLM to complete two user stories and two of their respective acceptance criteria. You must ensure that the generated user stories are correct and fix them if not. At the end of the submission, provide screenshots of your LLM prompts and answers, along with the description of errors the LLM made that you had to correct. Points will be deducted if no screenshots/description are provided.

For the other user stories and acceptance criteria, you are not required to use LLM.

I. User Stories (3 points)

- **User Story Template:** As a <role>, I want <goal> [so that <benefit>]

ID	User Story Name	User Story Description	Priority 1-5	Estimated effort (hours)
1	Choose a board size	As a player, I want to choose a board size so that I can customize the difficulty and enjoy a game of Peg Solitaire that matches my preferences.	1	3 hours
2	Choose the board type	As a Peg Solitaire player I want to pick the board type in the game so I can customize my experience.	3	3 hours

ID	User Story Name	User Story Description	Priority 1-5	Estimated effort (hours)
3	Start a new game of the chosen board size and type	As a Peg Solitaire player I want to get the game to the starting state of play, so that I can start to make moves in the game.	2	1 hour
4	Make a move in a game	As a Peg Solitaire player, I want to be able to interact with the board so I can play out a full game of Peg Solitaire	2	3-4 hours
5	A game is over	As a Peg Solitaire player want the game to be able to end so that I can play the game again	4	2-3 hours

II. Acceptance Criteria (AC) (12 points): Add/delete rows as needed.

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, InProgress)
1. Choose a board size	1.1	AC 1.1 Display available board sizes Given the player has opened the Peg Solitaire game, When the player accesses the board size selection menu, Then the game shall display the available board sizes for the player to choose from.	toDo
	1.2	AC 1.2 Select a board size Given the player is viewing the available board sizes, When the player selects a board size, Then the game shall save the selected board size as the configuration for the new game.	toDo
	1.3	AC 1.3 Confirm the selected board size Given the player has selected a board size, When the player proceeds to start a new game, Then the game shall use the selected board size to generate the game board	toDo
2. Choose a board type	2.1	AC 2.1 Display available board types Given the player has opened the Peg Solitaire game When the player accesses the board type selection menu Then the game shall display thr available board types for the play to choose from	toDo
	2.2	AC 2.2 Select a board type Given the player is viewing the available board types When the player selects a board type Then the game shall save the selected board type as the configuration for the new game.	toDo
	2.3	AC 2.3 Confirm the selected board type Given the player has selected a board type, When the player proceeds to start a new game Then the game shall use the selected board type to generate the game board.	toDo
3. Start a new game of the chosen board size and type	3.1	AC 3.1 Intitalize a game with the selected configuration Given the player has selected a valid board size and type to start the game When the game board is generated Then the board will be initialized with the specified positions and space based off the configuration	toDo
	3.2	AC 3.2 Display the game Given the player has started a game When the board initialization is complete Then the game will display the board to the player	toDo

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, InProgress)
4. Make a move in a game	4.1	AC 4.1 Make a valid move Given a game is in progress and the player selected a peg When the player selects a valid empty space that allows the peg to move. Then the game should move the peg to the empty space and remove the jumped over peg	toDo
	4.2	AC 4.2 prevent invalid move Given a game is in progress and the player selects a peg When the player attempts to make a move that violates the game rules Then the game should reject the move and not change the board	toDo
	4.3	AC 4.3 Allow additional moves Given the player has made a valid move and still has valid moves available When the board is updated Then the game should allow the player to move again	toDo
5. A game is over	5.1	AC 5.1 Detect a win state Given a game is in progress When the player reaches a specific win condition in the game Then the game shall state that the game has been won and prevent future moves	toDo
	5.2	AC 5.2 Detect a fail state Given a game is in progress When the player has no valid moves remaining and has not won Then the game will display a game over specifying the game is lost and prevent the player from moving.	toDo
	5.3	AC 5.3 Restart the game Given the game has been given a fail/win state When the player selects an option to start a new game Then a new game will initialize based off of the specified board types and sizes.	toDo

as part of a project involving a game of peg solitaire I need a user story that following the user story template As a <role>, I want <goal> [so that <benefit>]. Give me a description (that follows the previous format) of the user story that based off the following:
Choose a board size

User Story: Choose a Board Size



As a player, I want to choose a board size so that I can customize the difficulty and enjoy a game of Peg Solitaire that matches my preferences.



Choose the board type

User Story: Choose a Board Type



As a player, I want to choose a board type so that I can customize the layout of the Peg Solitaire game to match my preferences and enjoy different gameplay experiences.



Peg Solitaire – Acceptance Criteria

The following acceptance criteria cover all five user stories: choosing a board size, choosing a board type, starting a new game, making a move, and determining when a game is over.

Each criterion follows the **Given–When–Then** format.

1. Choose a Board Size

AC 1.1 – Display available board sizes

Given the player has opened the Peg Solitaire game,

When the player accesses the board size selection menu,

Then the game shall display the available board sizes for the player to choose from.

AC 1.2 – Select a board size

Given the player is viewing the available board sizes,

When the player selects a board size,

Then the game shall save the selected board size as the configuration for the new game.

AC 1.3 – Confirm the selected board size

Given the player has selected a board size,

When the player proceeds to start a new game,

Then the game shall use the selected board size to generate the game board.

2. Choose a Board Type

AC 2.1 – Display available board types

Given the player has opened the Peg Solitaire game,

When the player accesses the board type selection menu,

Then the game shall display the available board types for the player to choose from.

AC 2.2 – Select a board type

Given the player is viewing the available board types,

When the player selects a board type,

Then the game shall save the selected board type as the configuration for the new game.

AC 2.3 – Confirm the selected board type

Given the player has selected a board type,

When the player proceeds to start a new game,

Then the game shall use the selected board type to generate the game board.



A mistake that was made initially was that the LLM didn't provide an estimated amount of time needed to program the needed code for the first two user stories so I needed to request that specifically.

Today 11:26 AM

for the board size and board type user stories, how long do you think it would take an intermediate programmer to implement it in C++ in hours for each?

For an **intermediate C++ programmer** working on a Peg Solitaire project, assuming the GUI and basic game framework already exist, I'd estimate:

User Story	Estimated Time	Why
Choose a board size	2-4 hours	Add size-selection UI, store the selection, validate it, and generate the appropriate board dimensions.
Choose a board type	2-4 hours	Add type-selection UI, store the selection, and generate the appropriate board layout.

More specifically

Choose a board size — ~3 hours

- 30-45 min: Add board-size selection to the GUI
- 30 min: Create the C++ logic to store the selected size
- 1-1.5 hr: Generate/initialize the board based on the selected size
- 30 min: Testing and fixing edge cases

Choose a board type — ~3 hours

- 30-45 min: Add board-type selection to the GUI
- 30 min: Store and validate the selected type
- 1-1.5 hr: Implement the different board layouts
- 30 min: Testing